Monodromy in Radial Persistence Vineyards and visualizing with Bouquet2D

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DataShape Seminar, 14 October 2026











Structure

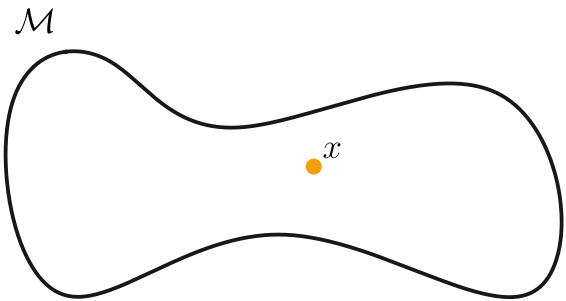
1. Radial Persistence Vineyards and Monodromy
2. Symmetry set and Focal set
3. The singular source of monodromy
4. Bouquet2D
5. Outlook

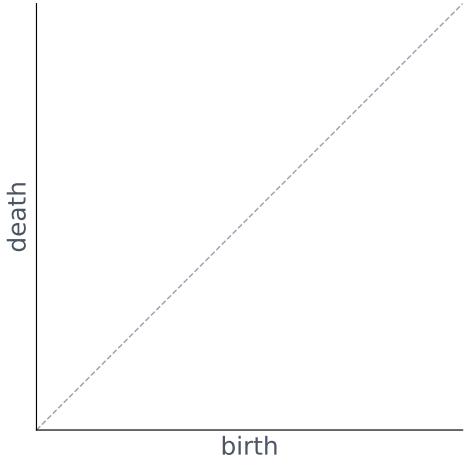
Radial persistence

Given a manifold $\mathcal{M} \subset \mathbb{R}^2$, a point $x \in \mathbb{R}^2$, the *radial persistence diagram* is given by the sublevel sets of the **squared Euclidean distance** function from x, restricted to the manifold $\mathcal{M}$:

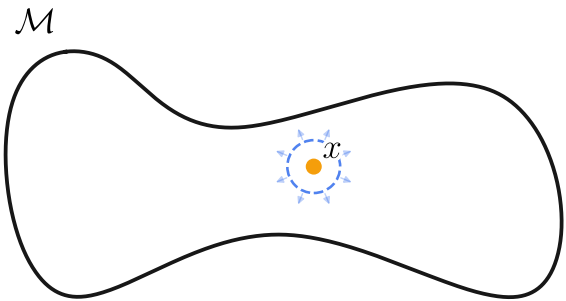
$$x \longmapsto \mathrm{Dgm}\big(d_E^2(\cdot, x)|_{\mathcal{M}}\big).$$

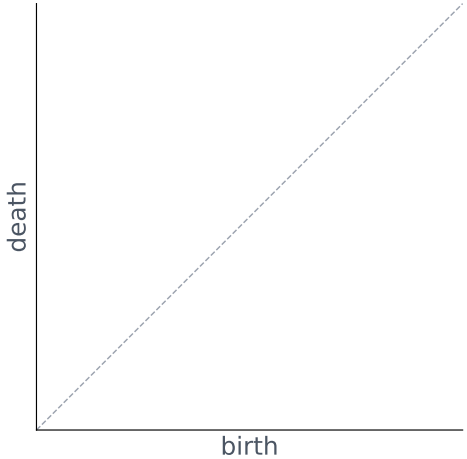
Radial persistence



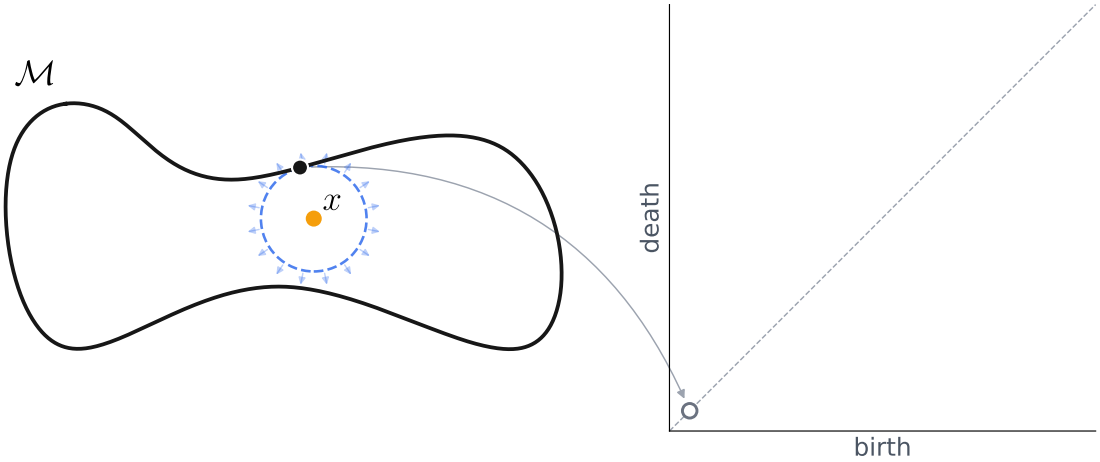


Radial persistence

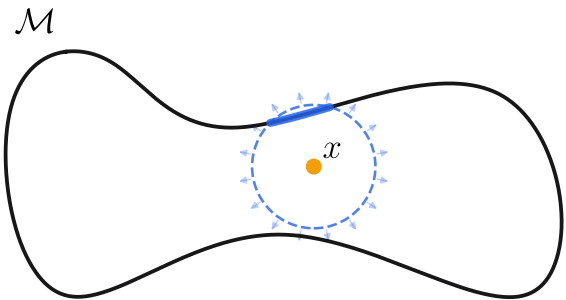


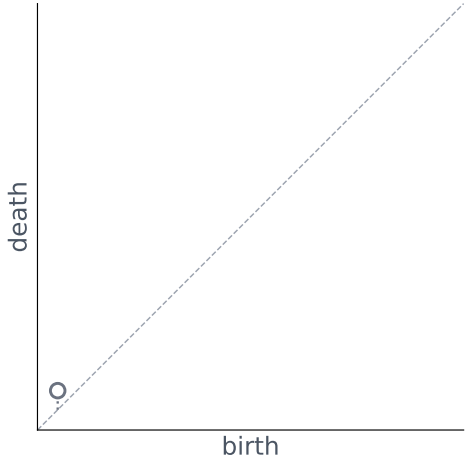


Radial persistence

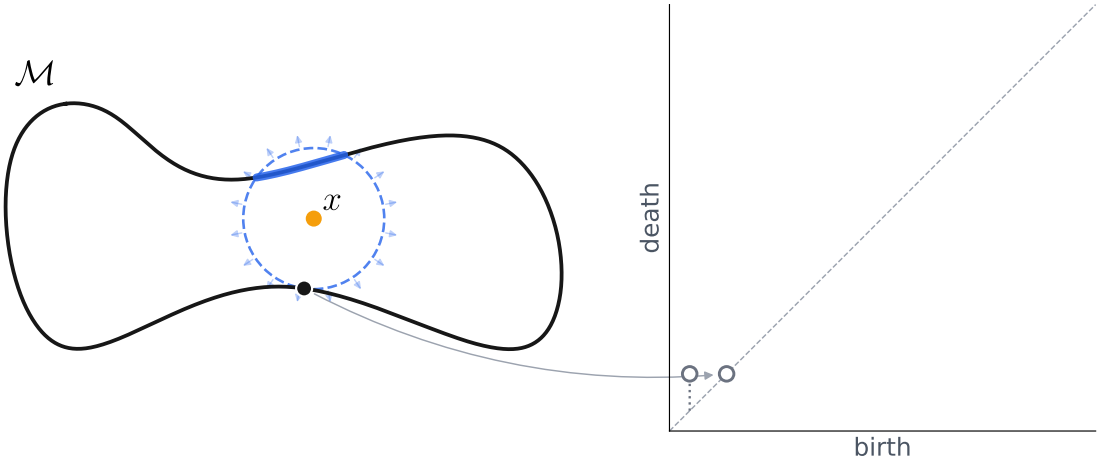


Radial persistence

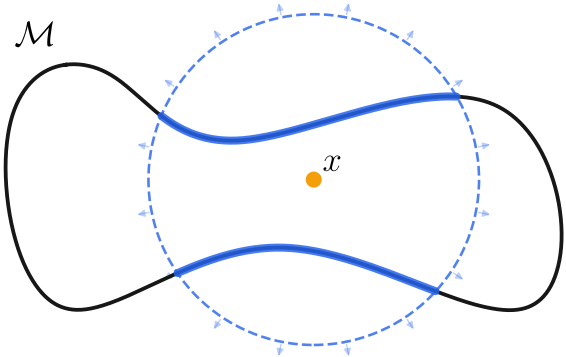


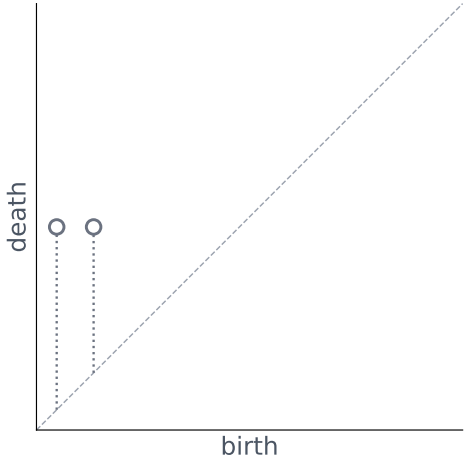


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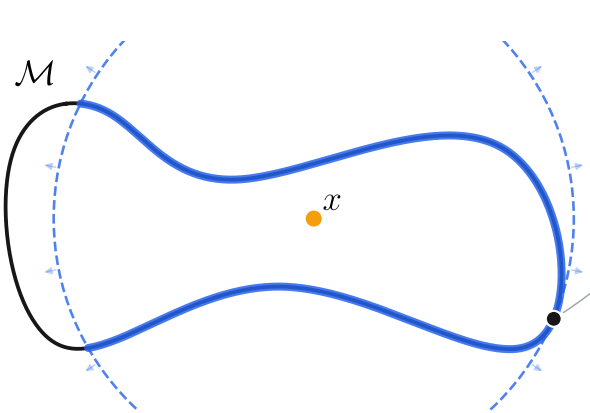


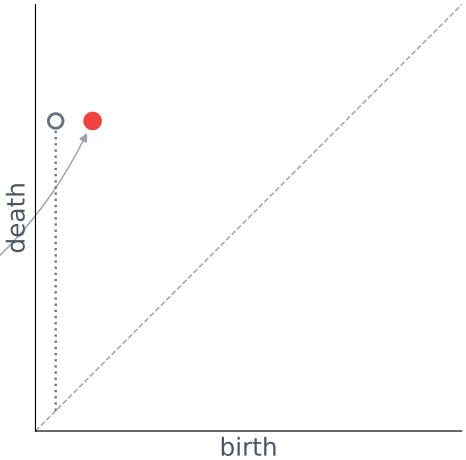
Radial persistence



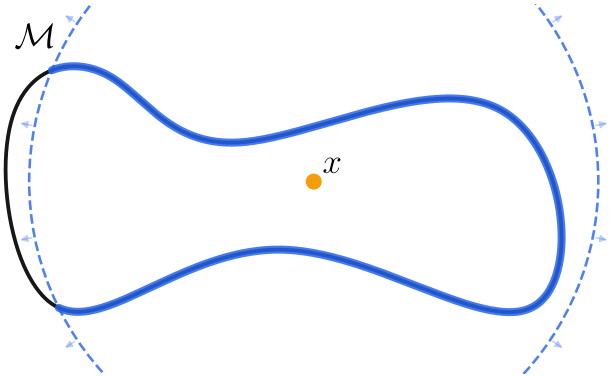


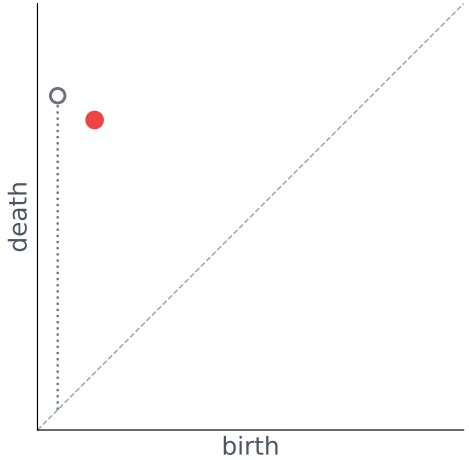
Radial persistence



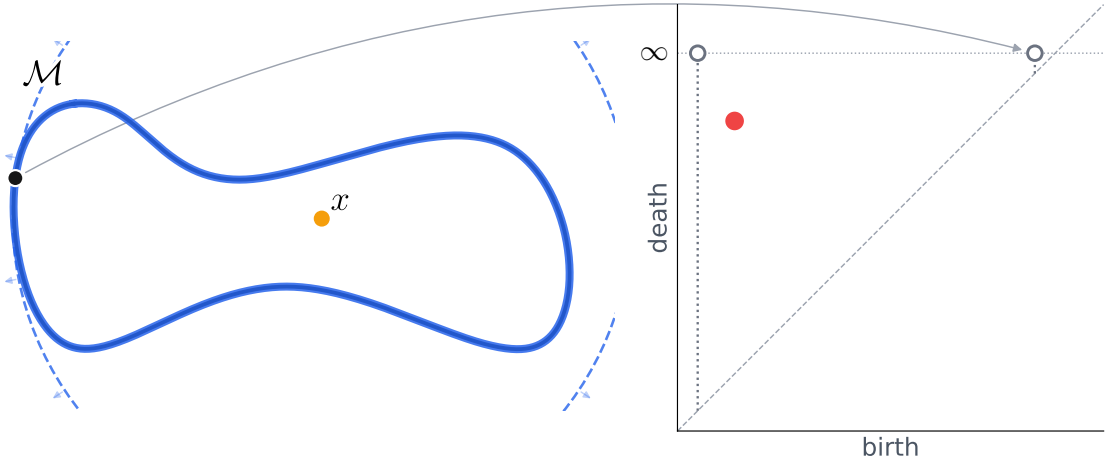


Radial persistence

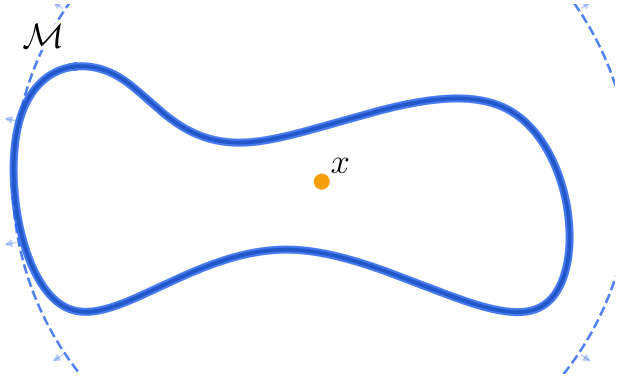


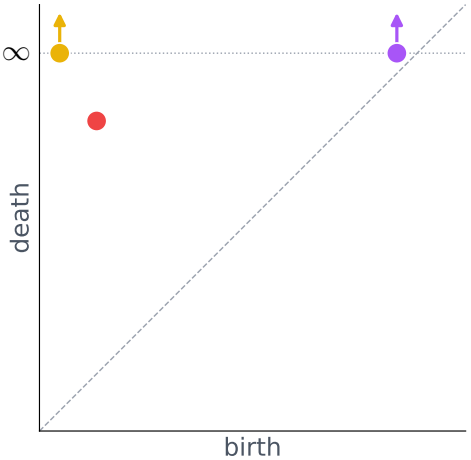


Radial persistence

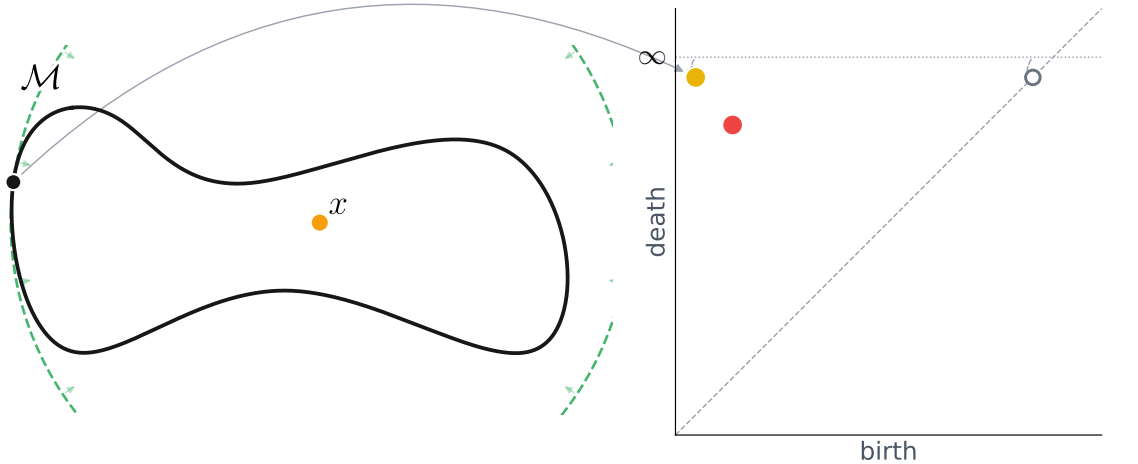


Radial persistence

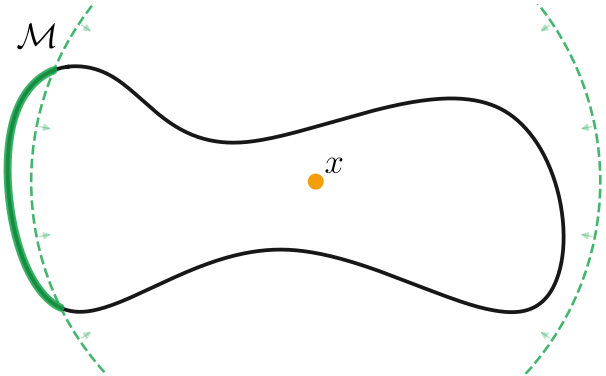


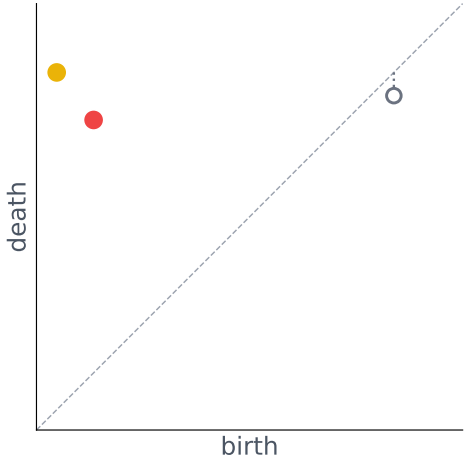


(Extended) Radial persistence

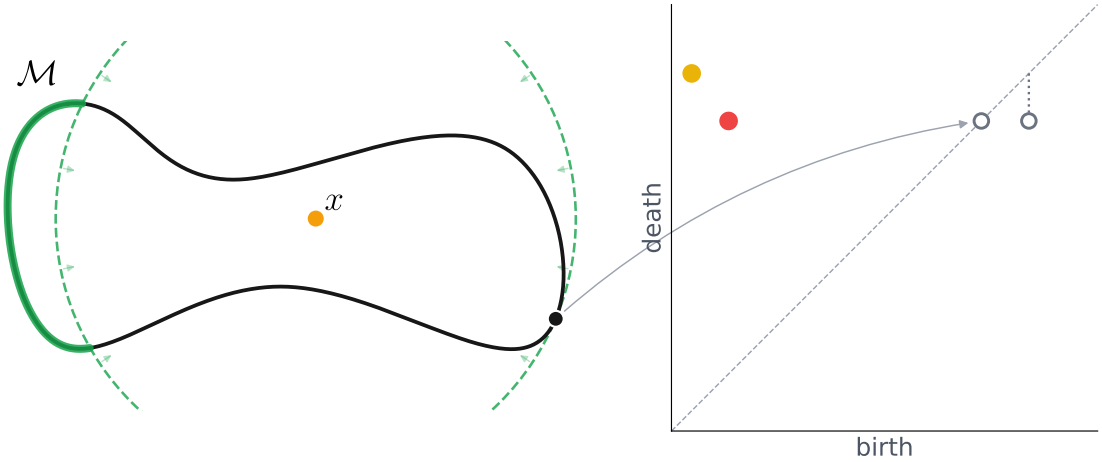


(Extended) Radial persistence

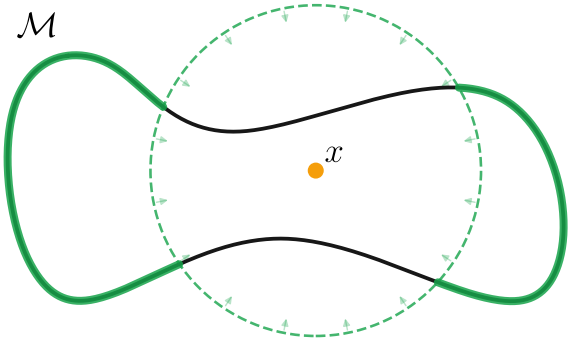


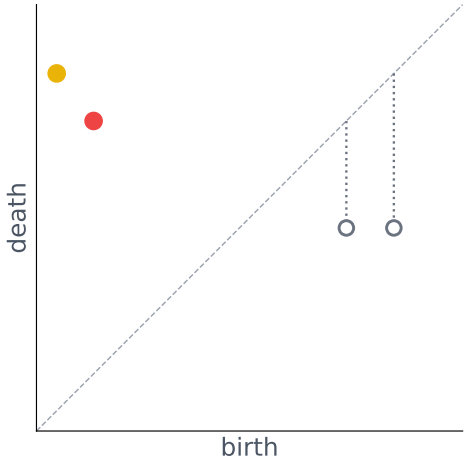


(Extended) Radial persistence

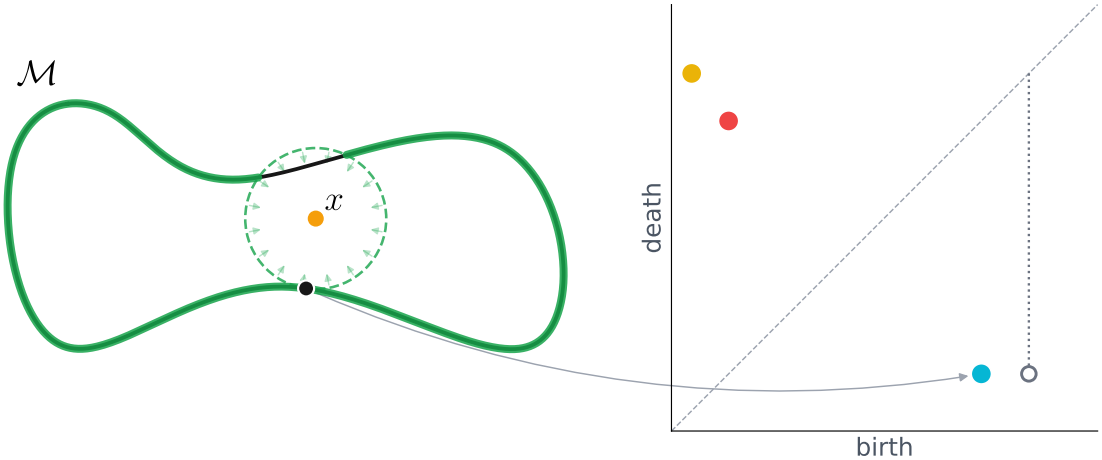


(Extended) Radial persistence

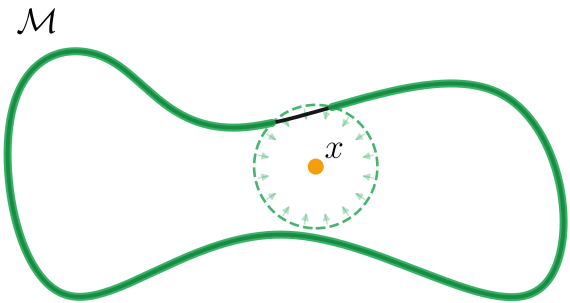


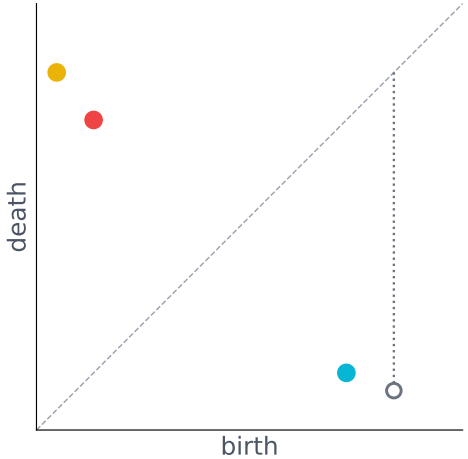


(Extended) Radial persistence

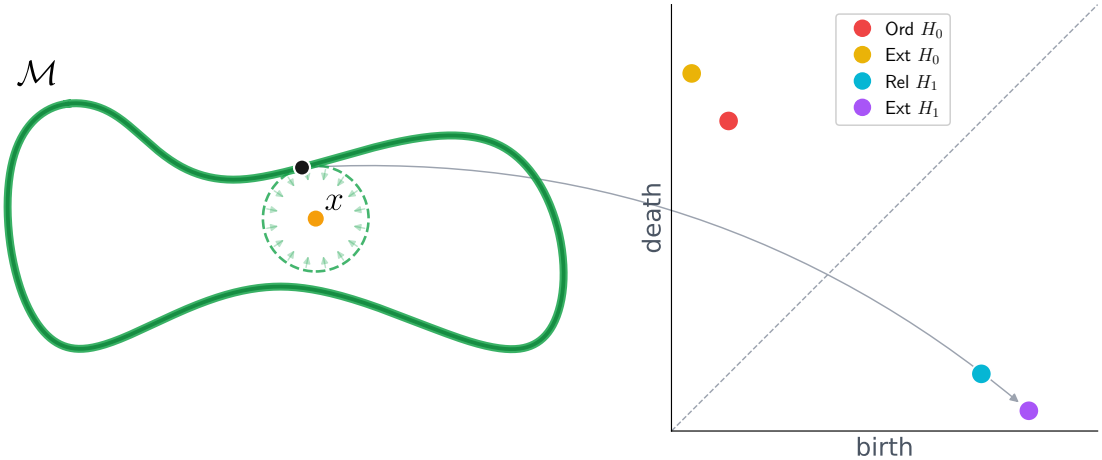


(Extended) Radial persistence

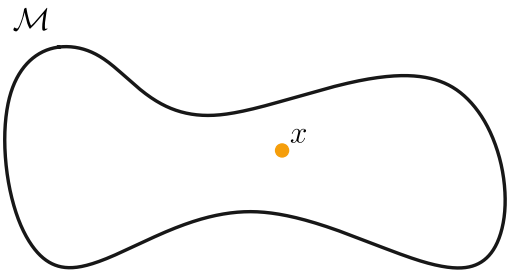


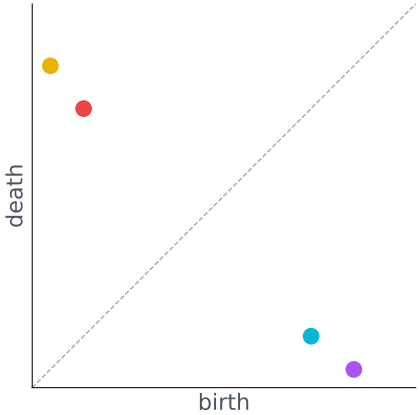


(Extended) Radial persistence

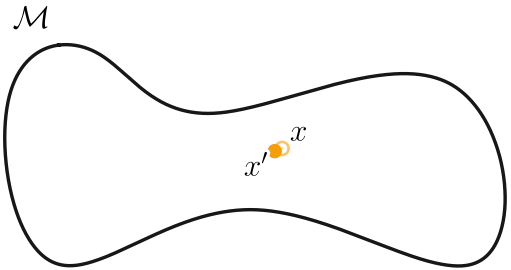


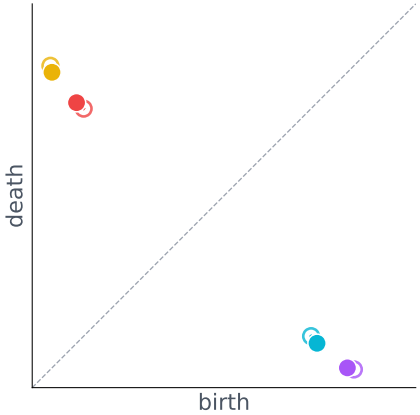
Stability



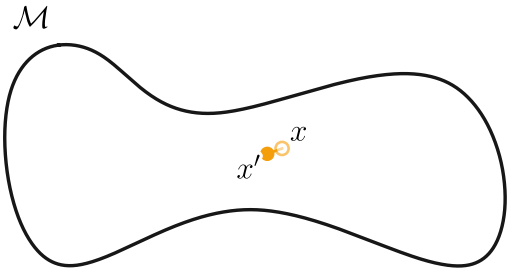


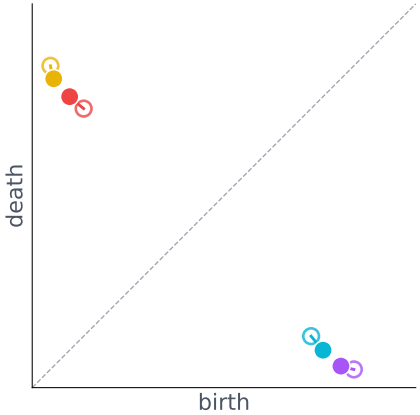
Stability



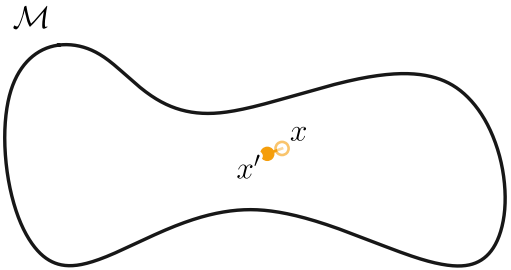


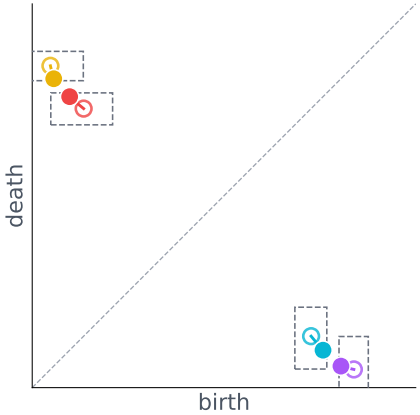
Stability





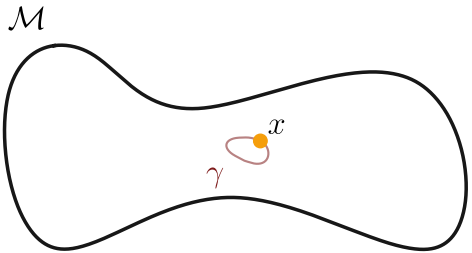
Stability

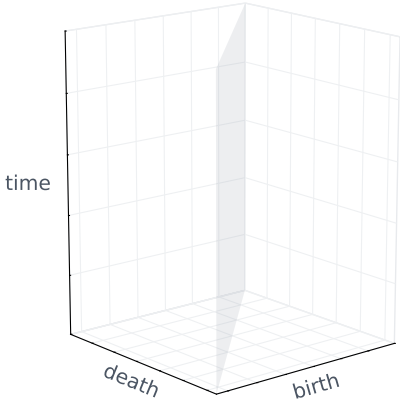




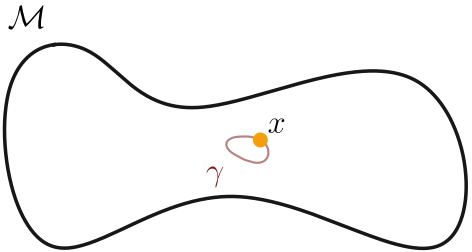
$$d_B\big(\mathrm{Dgm}(x), \mathrm{Dgm}(x')\big) \leq \|x - x'\|$$

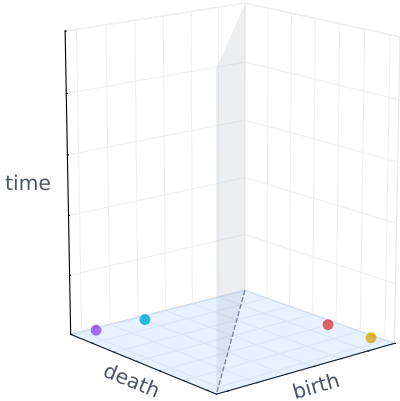
The Radial Persistence Transform



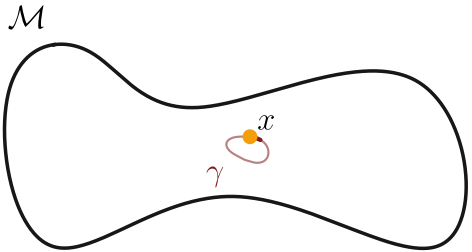


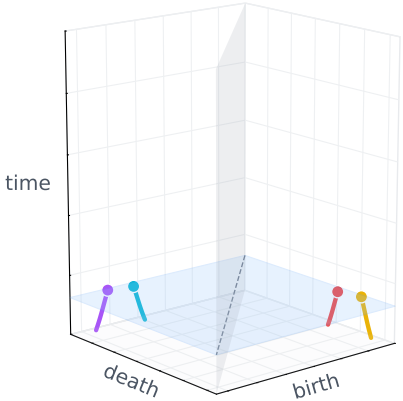
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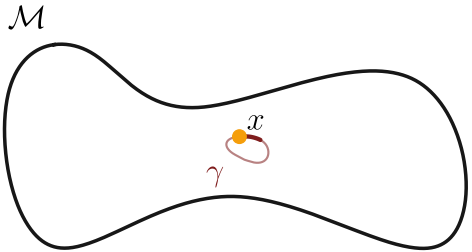


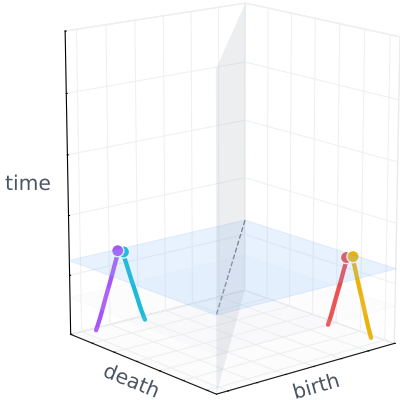
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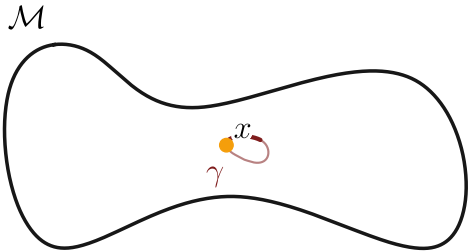


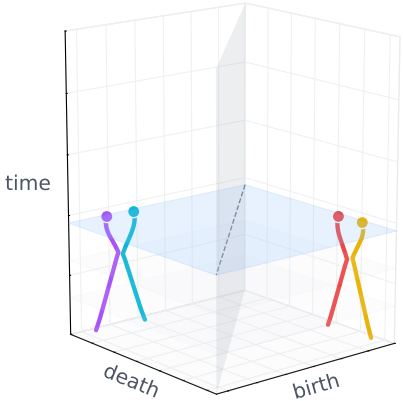
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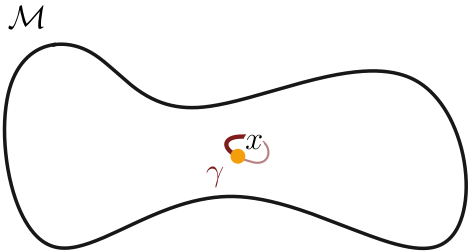


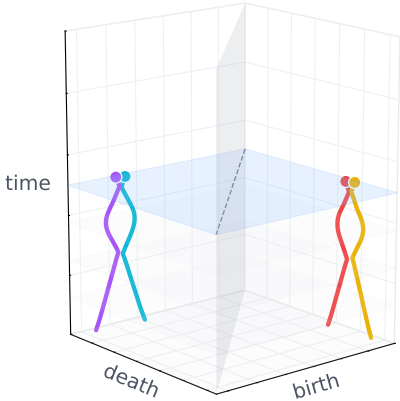
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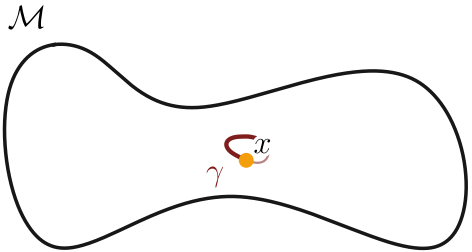


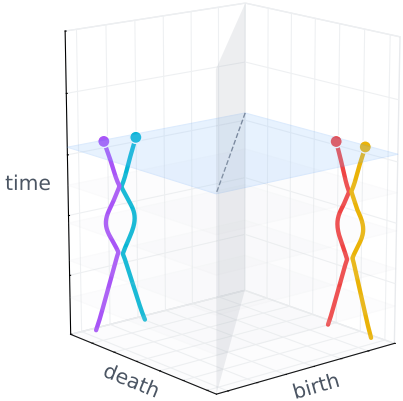
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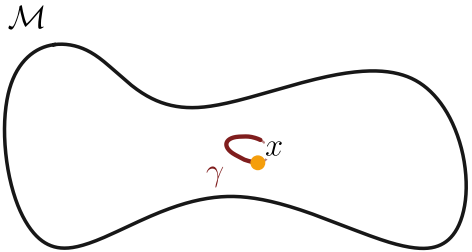


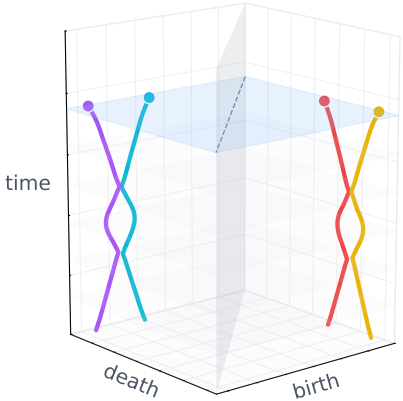
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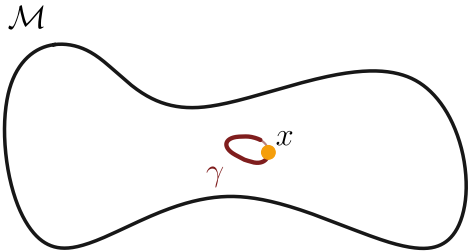


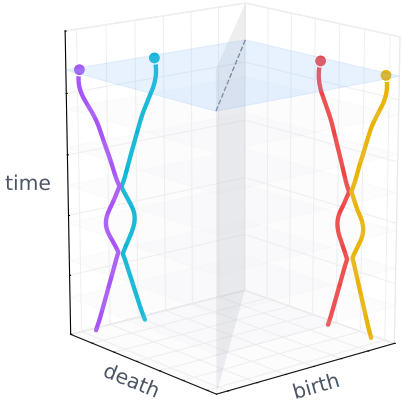
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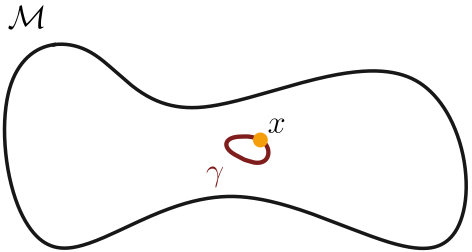


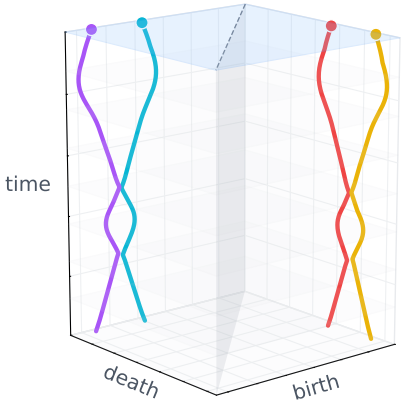
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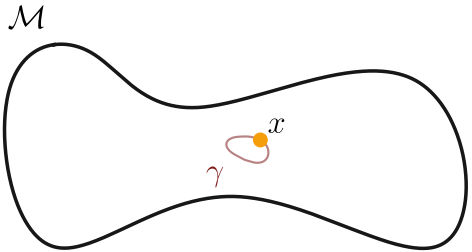


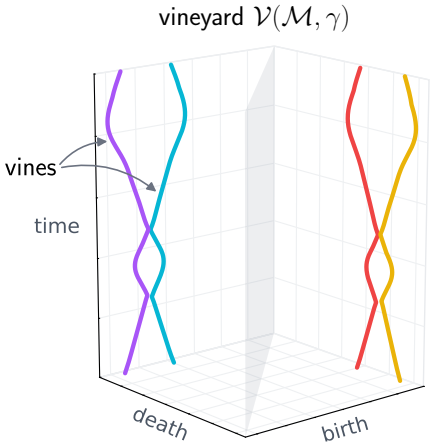
The Radial Persistence Transform





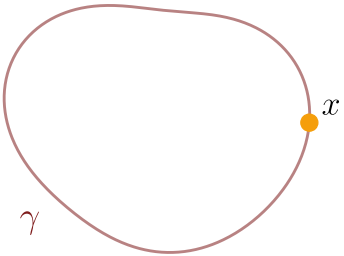
The Radial Persistence Transform

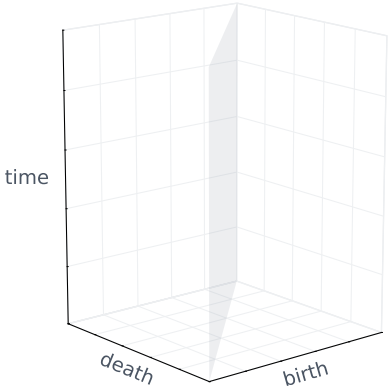




Monodromy

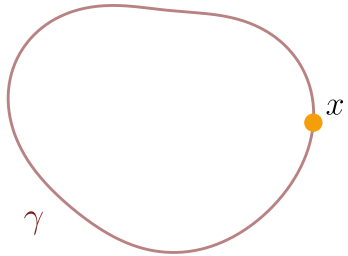
After one loop the diagram returns as a set, but its points are **permuted**.

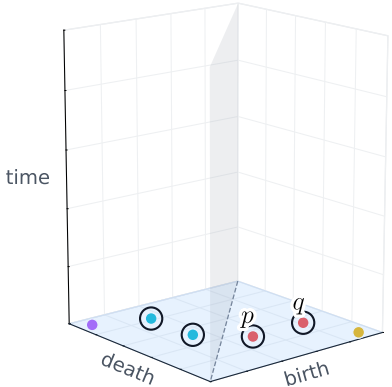




Monodromy

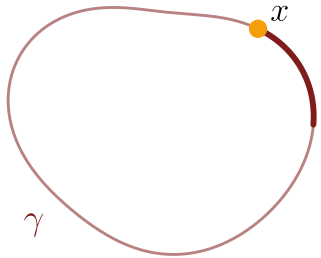
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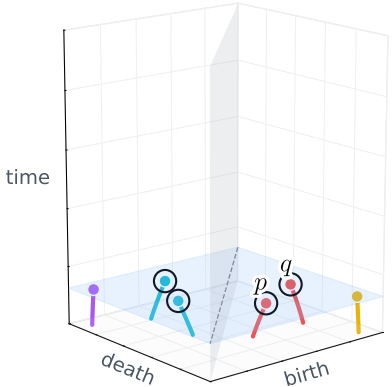




Monodromy

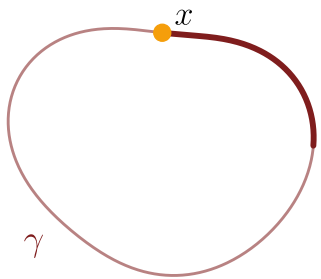
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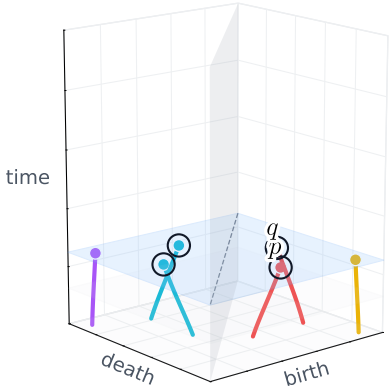




Monodromy

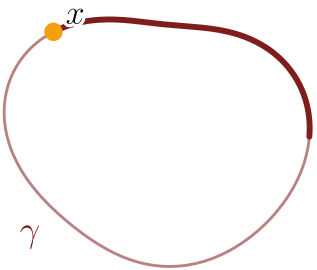
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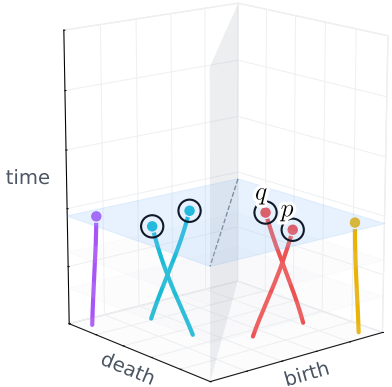




Monodromy

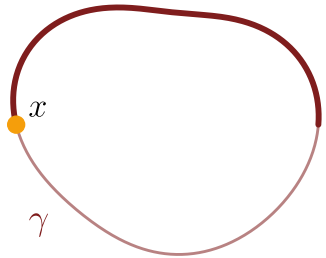
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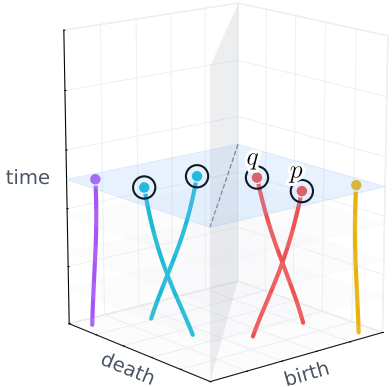




Monodromy

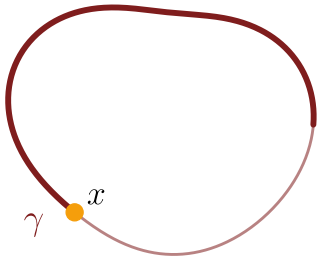
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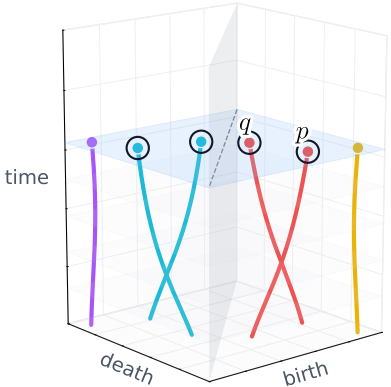




Monodromy

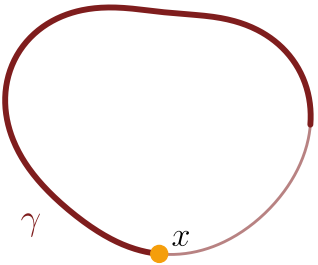
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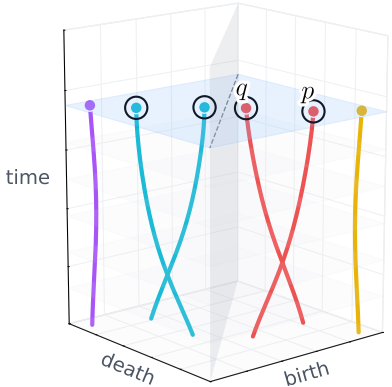




Monodromy

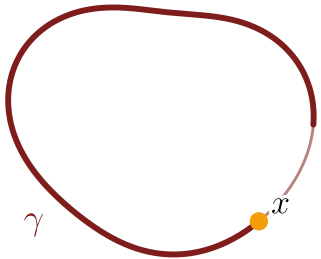
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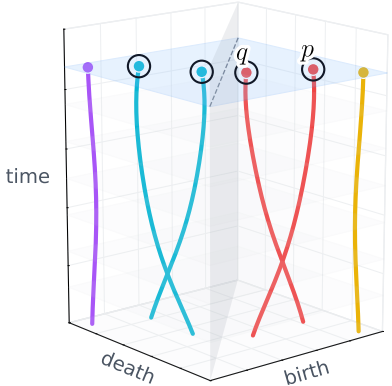




Monodromy

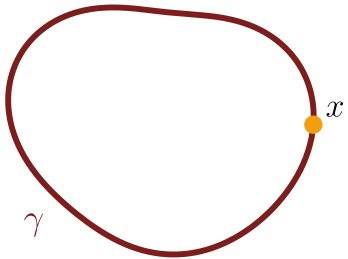
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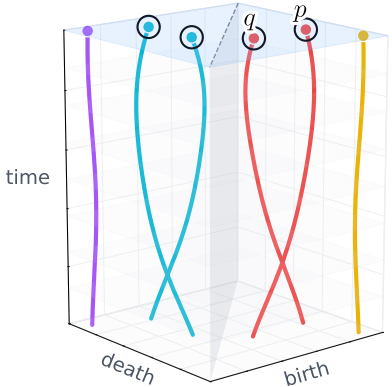




Monodromy

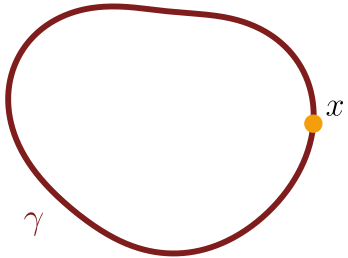
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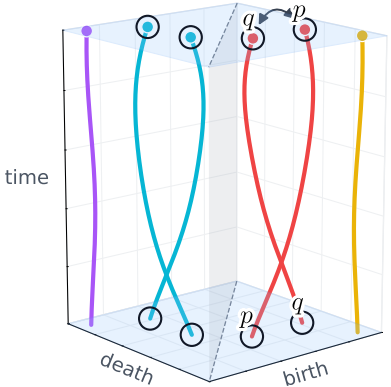


Monodromy

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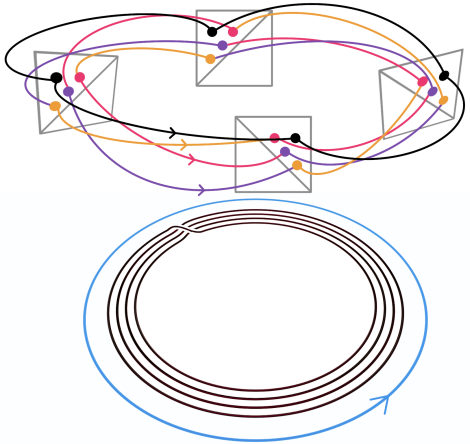


p and q are **permuted**



Order of Monodromy

The **order** of the monodromy is the smallest $k \geq 1$ such that following the vines k times around γ brings every point back to itself.



The driving question

Monodromy breaks feature tracking along γ.

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Which geometry of $\mathcal{M}$ **forces** monodromy?

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Answer: **one** singularity of the *symmetry set*.*

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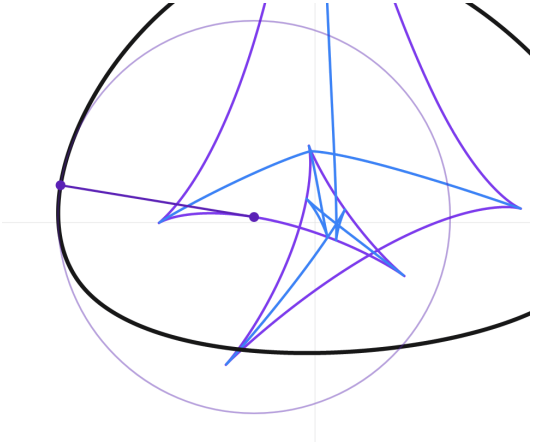
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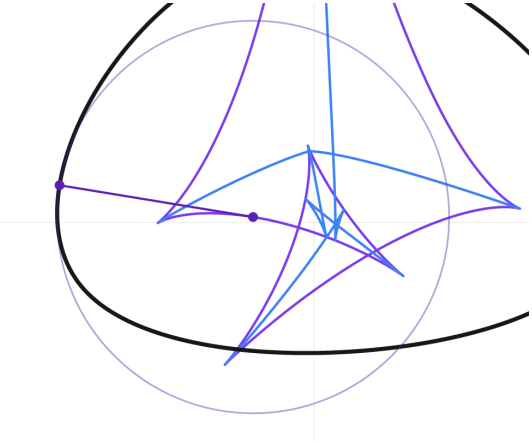
*Locally, in $\mathbb{R}^2$

Focal set



Set of centres of **osculating circles**.

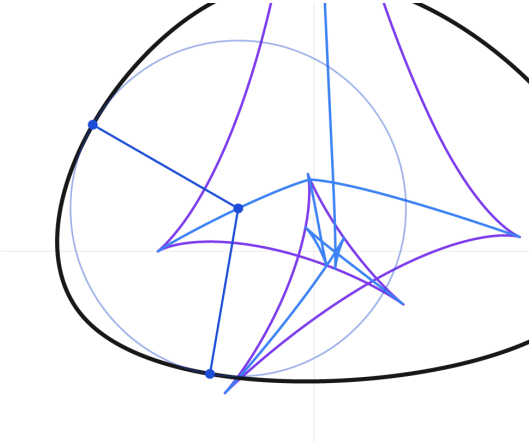
Focal set



Set of centres of **osculating circles**.

$x \in \mathrm{foc}(\mathcal{M}) \iff d_E^2(\cdot, x)|_{\mathcal{M}}$ has a **degenerate** critical point.

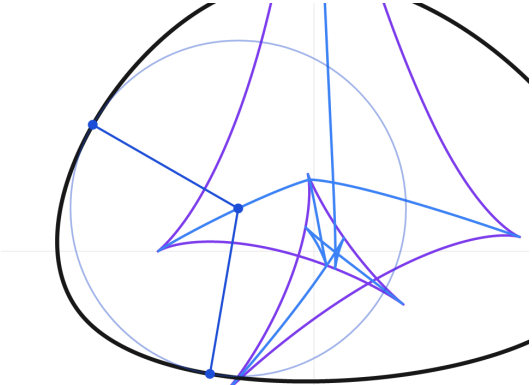
Symmetry set



Centres of circles **tangent to** $\mathcal{M}$ **at** ≥ 2 **points**.

Contains the **medial axis**.

Symmetry set

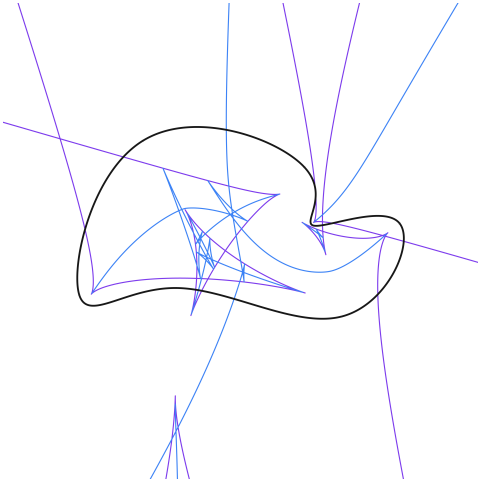


Centres of circles **tangent to** $\mathcal{M}$ **at** ≥ 2 **points**.

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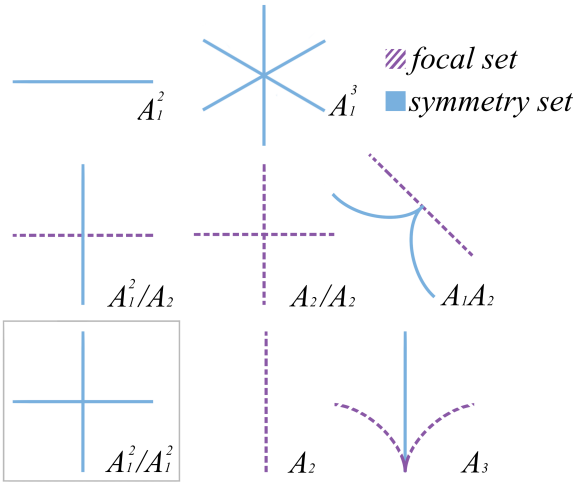
$x \in \mathrm{sym}(\mathcal{M}) \iff$ two critical **values** of $d_E^2(\cdot, x)|_{\mathcal{M}}$ coincide.

The bifurcation set



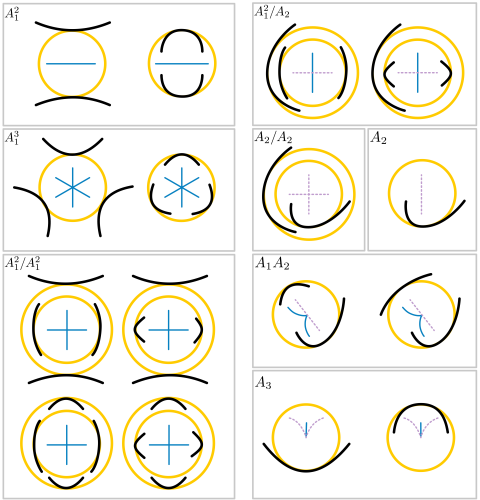
The **generalized symmetry set** $\mathrm{gsym}(\mathcal{M}) = \mathrm{sym}(\mathcal{M}) \cup \mathrm{foc}(\mathcal{M})$, also referred to as the **bifurcation set**.

Generic singularities in the plane

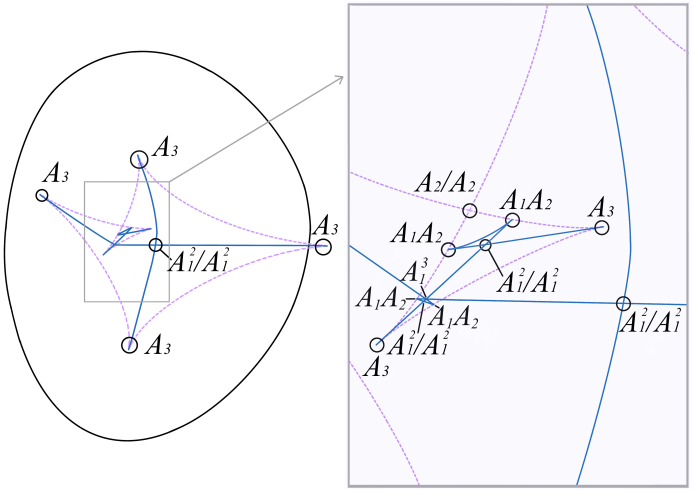


A_1^2	bitangent circle
A_1^3	triple tangency
A_1^2/A_2	concentric bitangent + osculating
A_2/A_2	two concentric osculating
A_1A_2	tangent + osculating
A_1^2/A_1^2	two concentric bitangents
A_2	osculating; focal
A_3	curvature extremum

Generic singularities in the plane



All of them on one curve

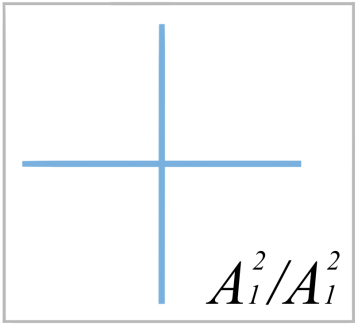


Main result

Theorem

Let $\mathcal{M} \subset \mathbb{R}^2$ be a generic smooth closed curve and let $\gamma : S^1 \to \mathbb{R}^2$ be a generic sufficiently small loop.

If γ encloses no singularity of type A_1^2/A_1^2, then $\mathcal{V}(\mathcal{M}, \gamma)$ has **no monodromy** and the vineyard consists of k **unlinked circles**.

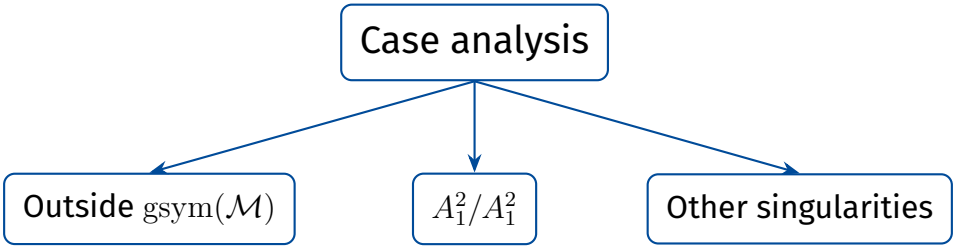


Proof?

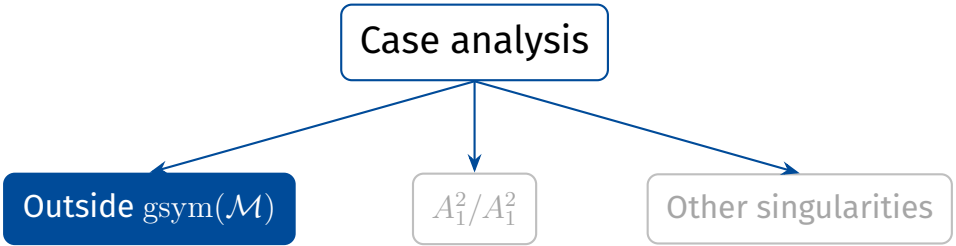
Proof?

Case analysis

Proof?

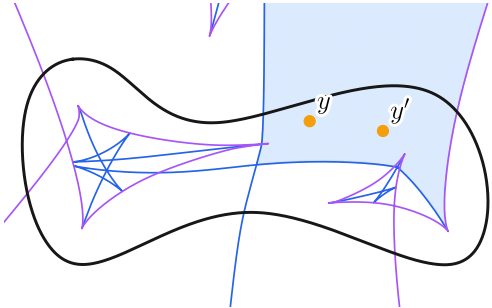

Case analysis
Outside $\mathrm{gsym}(\mathcal{M})$
A_1^2/A_1^2
Other singularities

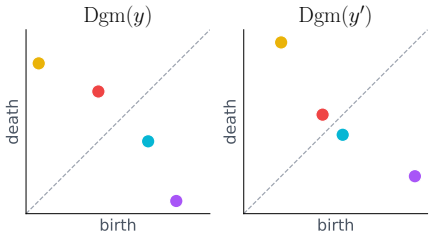
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Other singularities

Away from $\mathrm{gsym}(\mathcal{M})$ nothing happens

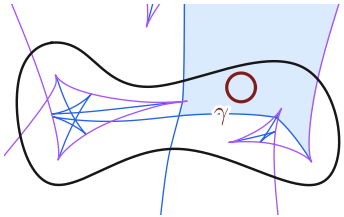
- On a component of $\mathbb{R}^2 \setminus \mathrm{gsym}(\mathcal{M})$ the diagrams are **combinatorially equivalent**.

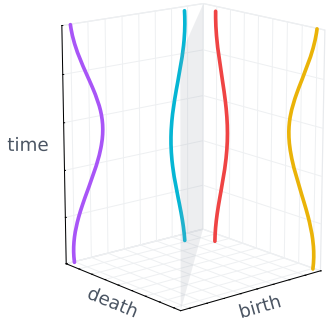


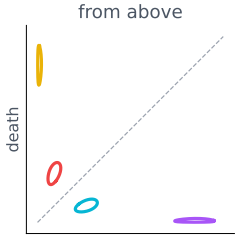


Away from $\mathrm{gsym}(\mathcal{M})$ nothing happens

- On a component of $\mathbb{R}^2 \setminus \mathrm{gsym}(\mathcal{M})$ the diagrams are **combinatorially equivalent**.
- $\Rightarrow$ a loop inside one component gives k **unlinked circles**.

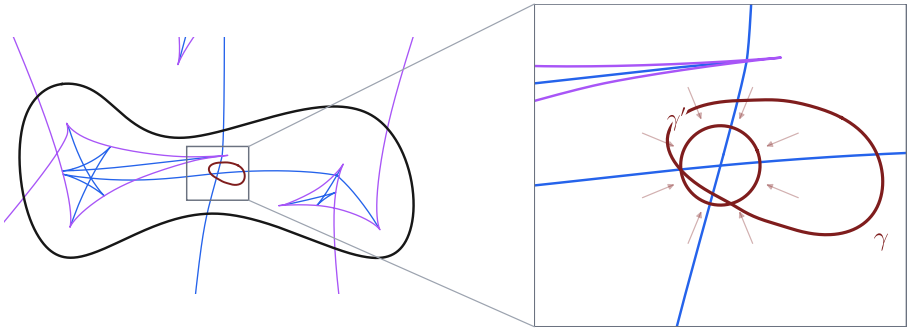


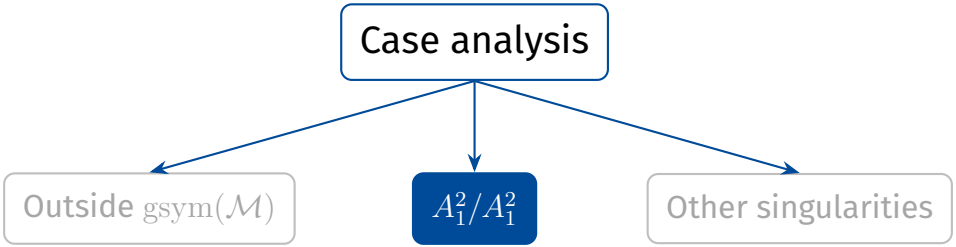




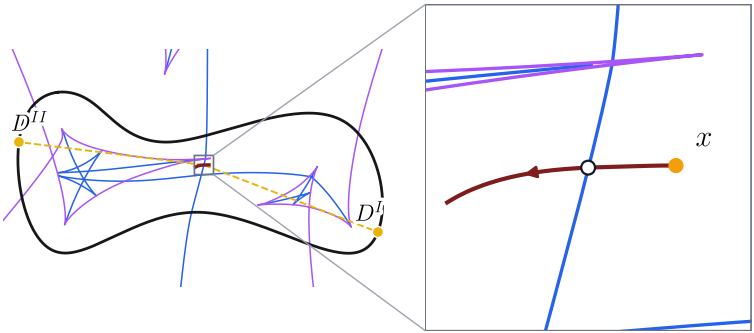
Away from $\mathrm{gsym}(\mathcal{M})$ nothing happens

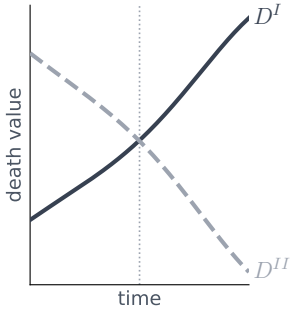
- On a component of $\mathbb{R}^2 \setminus \mathrm{gsym}(\mathcal{M})$ the diagrams are **combinatorially equivalent**.
- $\Rightarrow$ a loop inside one component gives k **unlinked circles**.
- $\Rightarrow$ only the *sequence* of components crossed matters: **loops can be shrunk**.




Case analysis
Outside $\mathrm{gsym}(\mathcal{M})$
A_1^2/A_1^2
Other singularities

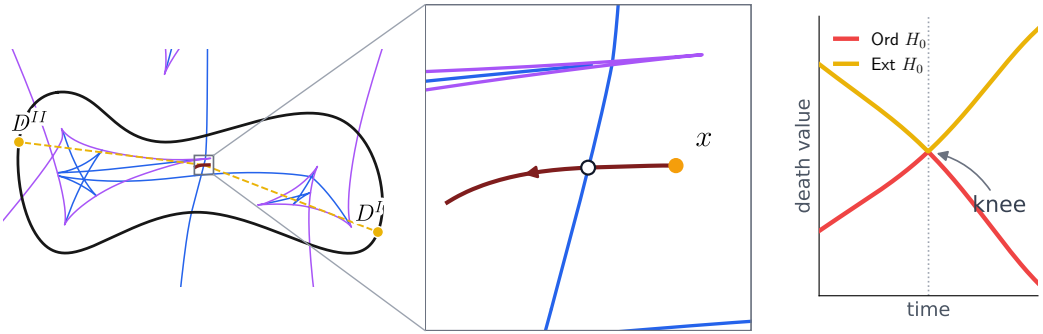
Crossing the symmetry set (A_1^2)





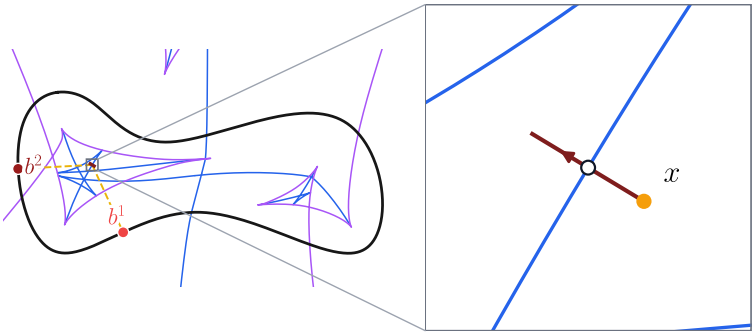
Two critical values **swap order**.

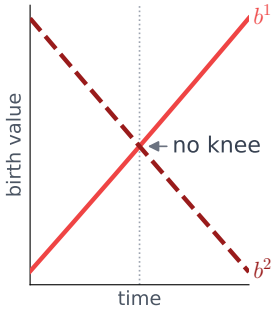
Crossing the symmetry set (A_1^2)



Two critical values **swap order**. The elder rule may re-pair them: **knees**.

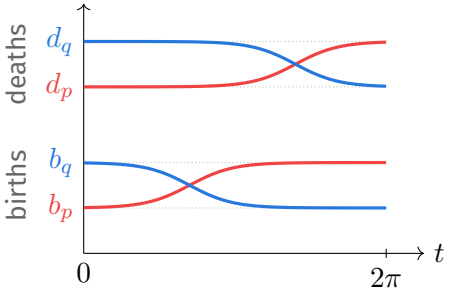
Crossing the symmetry set without the elder rule





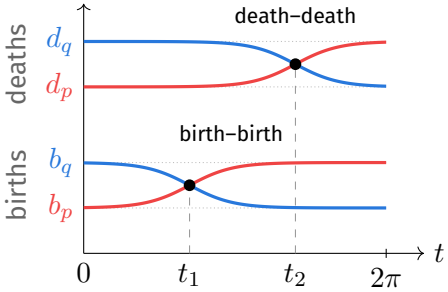
Two births **swap order**, but each keeps its own death: the pairing does not change, **no knee**.

What an interchange needs



After one loop p and q have **swapped**: p ends at (b_q, d_q), q at (b_p, d_p).

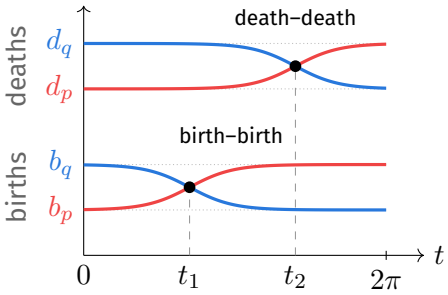
What an interchange needs



After one loop p and q have **swapped**: p ends at (b_q, d_q), q at (b_p, d_p).

$\Rightarrow$ their births cross at some t_1, their deaths at some t_2: $\gamma(t_1)$ is on a **birth–birth** and $\gamma(t_2)$ on a **death–death** stratum of $\mathrm{sym}(\mathcal{M})$.

What an interchange needs

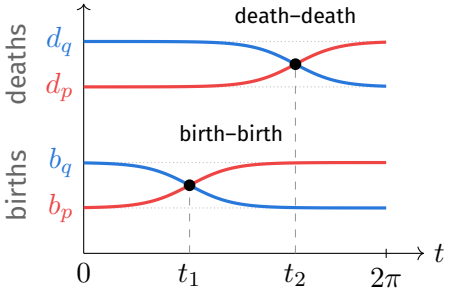


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$\Rightarrow$ at least **4 contact points**, from exactly **two** persistence pairs.

What an interchange needs



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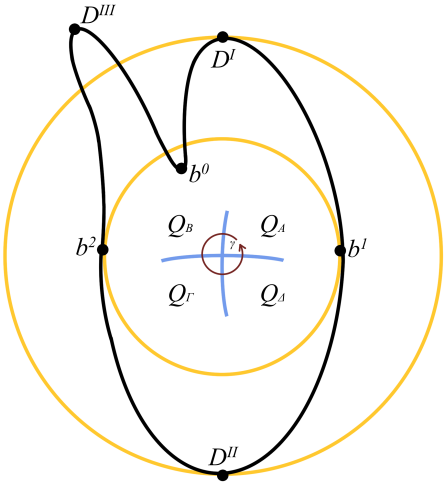
$\Rightarrow$ their births cross at some t_1, their deaths at some t_2: $\gamma(t_1)$ is on a **birth–birth** and $\gamma(t_2)$ on a **death–death** stratum of $\mathrm{sym}(\mathcal{M})$.

$\Rightarrow$ at least **4 contact points**, from exactly **two** persistence pairs.

Consequence

Only A_1^2/A_1^2 has 4 such contact points:
it is the **only singularity that can cause an interchange**.

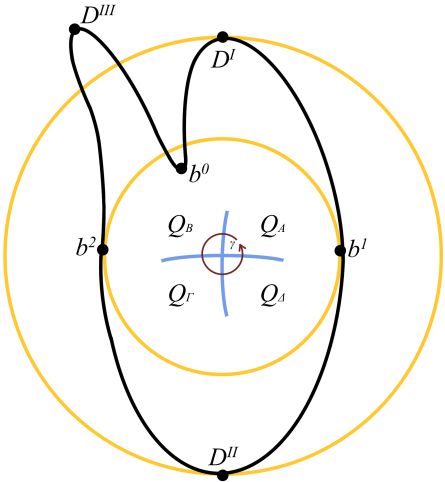
The culprit: A_1^2/A_1^2



b^0 is born first and D^{III} dies last.

Quadrant	pairing
Q_A	(b^1, D^I), (b^2, D^{II})

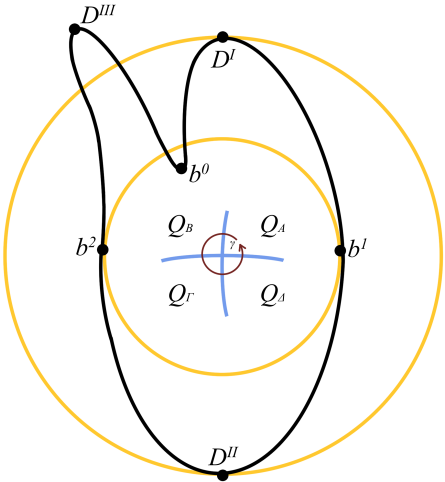
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b^0 is born first and D^{III} dies last.

Quadrant	pairing
Q_A	(b^1, D^I), (b^2, D^{II})
Q_B	(b^1, D^I), (b^2, D^{II})

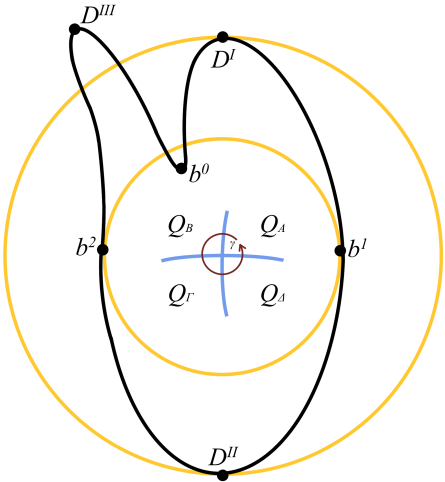
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Quadrant	pairing
Q_A	(b^1, D^I), (b^2, D^{II})
Q_B	(b^1, D^I), (b^2, D^{II})
Q_Γ	(b^2, D^I), (b^1, D^{II})

The culprit: A_1^2/A_1^2

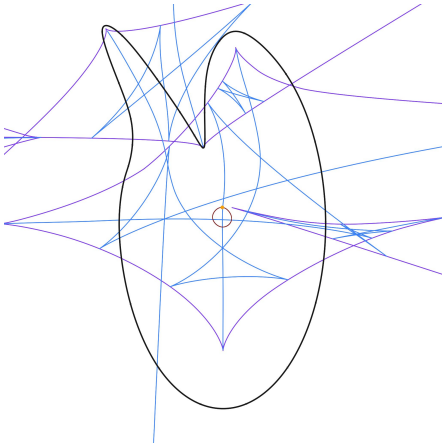


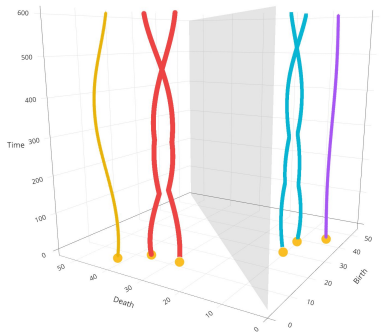
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Quadrant	pairing
Q_A	(b^1, D^I), (b^2, D^{II})
Q_B	(b^1, D^I), (b^2, D^{II})
Q_Γ	(b^2, D^I), (b^1, D^{II})
Q_Δ	(b^1, D^I), (b^2, D^{II})

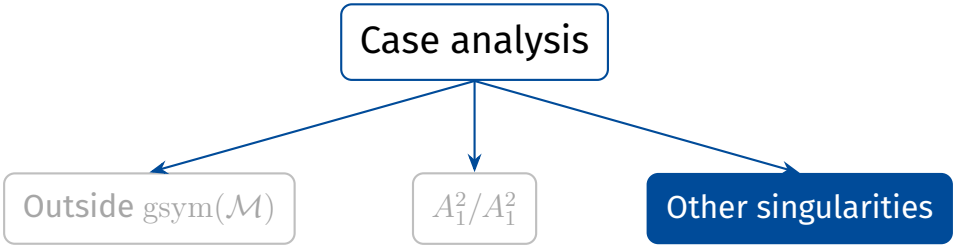
Two consecutive changes ($B \to \Gamma \to \Delta$).

The resulting vineyard

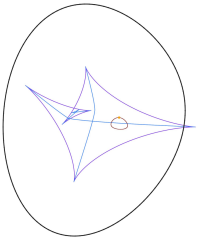


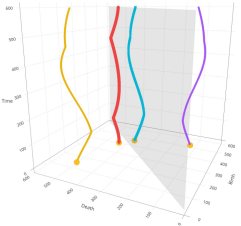


The red and turquoise pairs **interchange** after one loop.

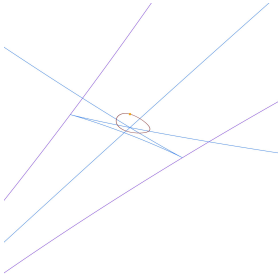

Case analysis
Outside $\mathrm{gsym}(\mathcal{M})$
A_1^2/A_1^2
Other singularities

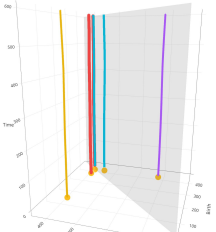
Symmetry set only: A_1^2 and A_1^3





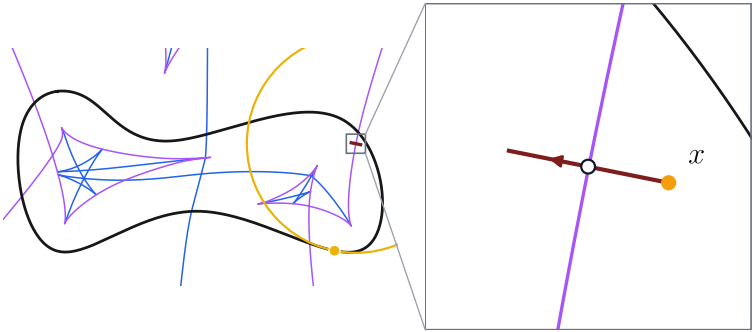
A_1^2: the same stratum is crossed twice, in opposite directions. Everything is undone.

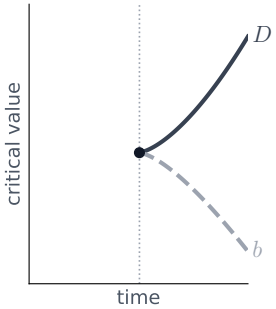




A_1^3: only **3** contact points. No interchange; points stay in disjoint zones.

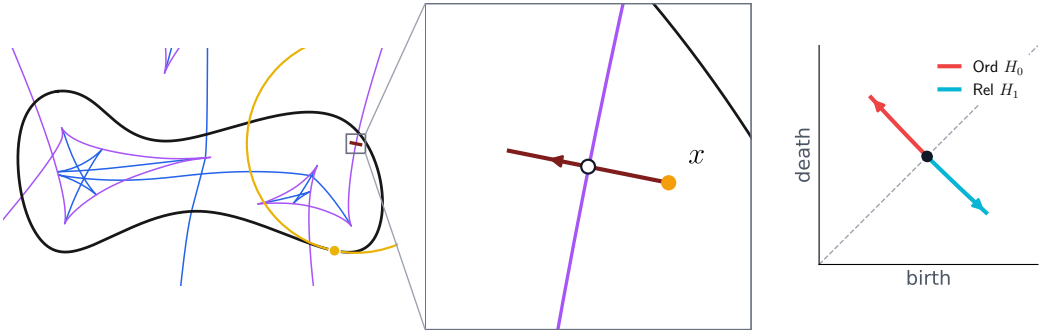
Crossing the focal set (A_2)





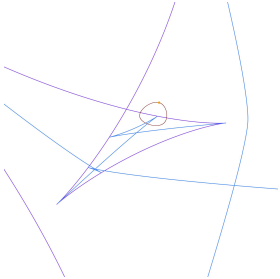
A new birth and death **appear together**.

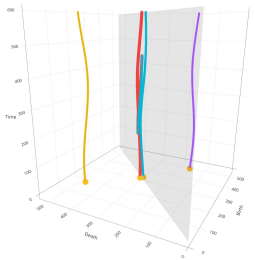
Crossing the focal set (A_2)



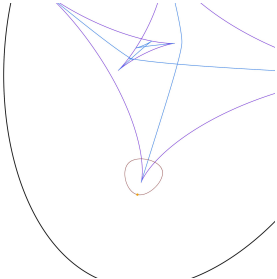
A new birth and death **appear together**. The pair is **born from** the diagonal.

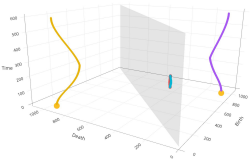
Focal set involved: short vines





A_1A_2: a point pops out of the diagonal; it is too short to interchange.

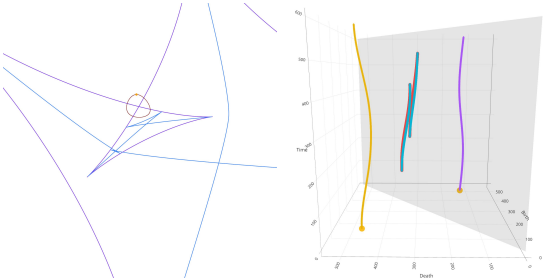




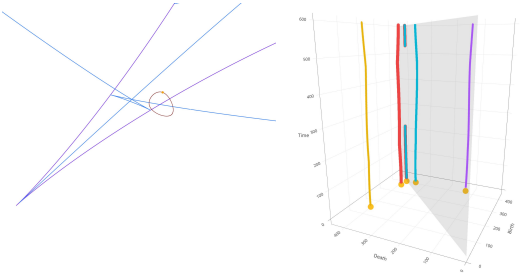
A_3: the same, with an elder-rule knee in between.

Small loop $\Rightarrow$ vines from the diagonal have the **least persistence**: by stability they cannot link with the rest.

Focal set involved: short vines

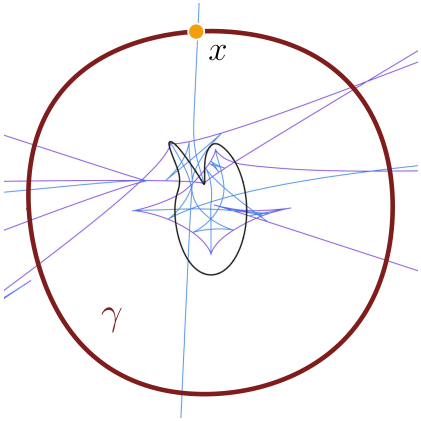


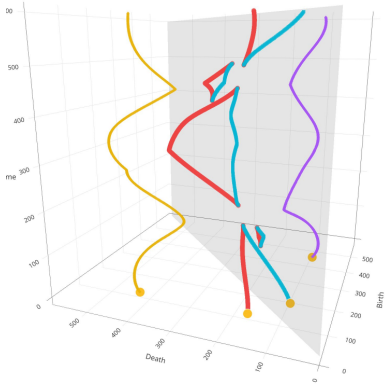
A_2/A_2: two short vines, at distinct radii $r_1 \neq r_2$.



A_1^2/A_2: a short vine, and only 2 contacts on the bitangent circle.

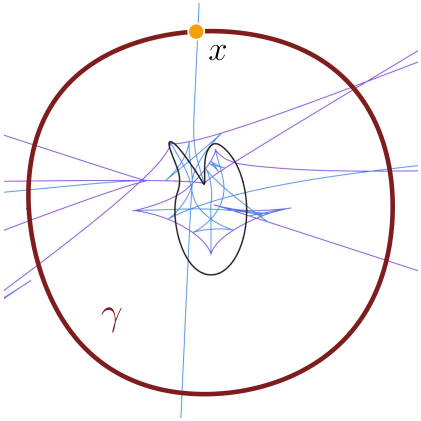
Local $\neq$ global

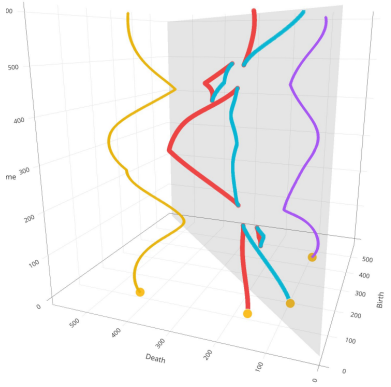




The converse fails: a **big** loop around the same A_1^2/A_1^2 has **no** monodromy.

Local $\neq$ global

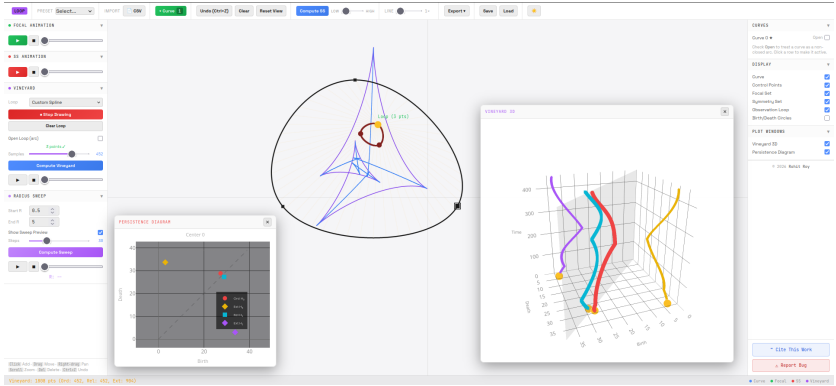




The converse fails: a **big** loop around the same A_1^2/A_1^2 has **no** monodromy.

Ongoing: a **global** classification.

Bouquet2D



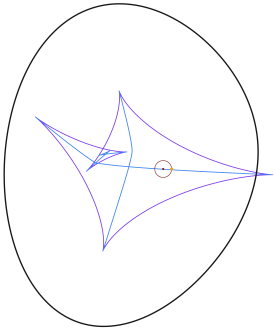
Interactive, in the browser.

bouquet2d.rohitroy.me

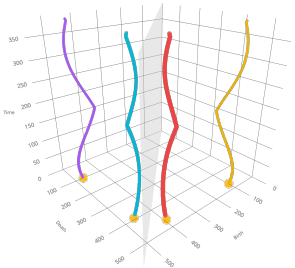
What Bouquet2D does

Input: a spline curve $\mathcal{M}$ and a loop γ (circle or hand-drawn).

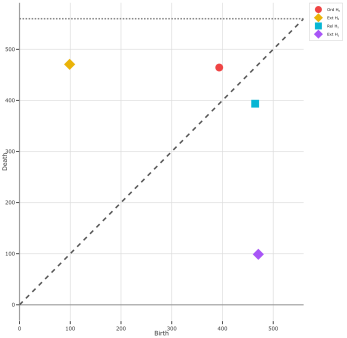
Output, in real time:



symmetry & focal sets



the 3D vineyard



the extended diagram at $\gamma(t)$

Demo



https://bouquet2d.rohitroy.me

Limitations

- **Planar only:** curves in $\mathbb{R}^2$.
- **Sampling-sensitive:** the discrete symmetry set can miss or add branches near singularities.
- **Threshold-sensitive:** near singularities, results depend on user-set constants.

Summary and open questions

Summary

Small loops in $\mathbb{R}^2$: monodromy $\Rightarrow$ an enclosed A_1^2/A_1^2 point of the symmetry set.

- **Global** criteria: how do singularities combine along big loops?
- **Higher dimensions**: Bruce–Giblin–Gibson also classify $\mathbb{R}^3$. (Bouquet3D?)
- **Examples** of topology and monodromy in vineyards from data.

Thank You!

Questions?

Paper: arXiv:2607.01046 Bouquet2D: `https://bouquet2d.rohitroy.me`

References

[1] Turner, Mukherjee & Boyer, *Persistent homology transform for modeling shapes and surfaces*, Inf. Inference 2014.

[2] Onus, Otter & Turkes, *Shoving tubes through shapes gives a sufficient and efficient shape statistic*, arXiv:2412.18452, 2024.

[3] Cohen-Steiner, Edelsbrunner & Morozov, *Vines and vineyards by updating persistence in linear time*, SoCG 2006.

[4] Turner, *Representing vineyard modules*, arXiv:2307.06020, 2023.

[5] Arya, Giunti, Hickok, Kanari, McGuire & Turner, *Decomposing the persistent homology transform of star-shaped objects*, arXiv:2408.14995, 2024.

[6] Chambers, Fillmore, Stephenson & Wintraecken, *Braiding vineyards*, SODA 2026.

[7] Bruce, Giblin & Gibson, *Symmetry sets*, Proc. Roy. Soc. Edinburgh A, 1985.

[8] Giblin & Diatta, *Symmetry sets and medial axes in two dimensions*, 2003.

[9] Porteous, *Geometric differentiation*, Cambridge University Press, 2001.

[10] Edelsbrunner, Stephenson & Thoresen: *The mid-sphere cousin of the medial axis transform*, 4th International Joint Conference on Discrete Geometry and Mathematical Morphology, 2025.

[11] Chambers, Fillmore, Mukherjee, Roy, Stephenson & Wintraecken, *Bouquet: a visualization tool for symmetry sets and vineyards*, EuroCG 2026.

[12] Chambers, Fillmore, Mukherjee, Roy, Stephenson & Wintraecken, *The singular source of vineyard monodromy*, arXiv:2607.01046, 2026.